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Unlocking Potential for Mauddud Carbonate Formation in the Greater Burgan Field, Kuwait

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Abstract

The primary objective of this paper is to introduce fit-for-purpose technologies and best practices aimed at increasing and sustaining the productivity of the Mauddud reservoir in the Greater Burgan Field, Kuwait, to a level of 40,000 bbl/day by 2030. Additional objectives include constructing DFN and 3D geomechanics models, and developing a mechanistic reservoir model to better understand reservoir fluid flow and select sweet spots for new horizontal infill wells.

The approach involved a comprehensive analysis of the Mauddud Formation, characterized by thin layers, a tight carbonate matrix, and low productivity. The process included data gap analysis, updating the static model, and constructing 3D and 1D MEM geomechanics models. Natural fracture modeling was employed to understand the distribution of matrix permeability and fracture networks. Machine learning was used to propose development strategies, and cost-effective technology management programs were employed for drilling and workover candidate wells.

The study reveals that natural fractures in the Mauddud formation are oriented NE-SW and are fault-related, formed due to multiple tectonic episodes (Fig. 2). The recovery mechanisms are flow from natural fractures and acid frac jobs supported by a larger volume of low-matrix permeability. Single porosity models were found to be inadequate to reproduce production behavior. Wells located within 500 meters of the main faults showed better productivity. Implementing multi-stage acid fracturing in two newly horizontal wells drilled based on study recommendations has increased current oil production by 60%.

This paper presents novel insights into the Mauddud Formation's fracture orientation and recovery mechanisms, offering a new understanding of the reservoir's behavior. The use of advanced modeling techniques and machine learning to propose development strategies provides valuable contributions to the existing body of literature, benefiting the petroleum industry by enhancing reservoir management and production optimization.

Geological & Reservoir Background

The Greater Burgan Field, located in southeastern Kuwait, is one of the largest oil accumulations globally, with multiple stacked reservoirs ranging from Lower Cretaceous to Tertiary age. Within this complex, the Mauddud Formation represents a Lower Cretaceous carbonate unit deposited in a shallow marine ramp setting. The reservoir is characterized by laterally continuous, thin-bedded limestone and dolomitic intervals, interbedded with marls and argillaceous layers that act as intra-formational barriers.

The Mauddud carbonate matrix exhibits inherently low permeability, typically in the range of tens of millidarcies or less, and moderate porosity averaging 15-20%. The reservoir quality is strongly influenced by diagenetic processes, including cementation, dissolution, and dolomitization, which create heterogeneity at multiple scales. Natural fractures play a critical role in enhancing effective permeability, particularly in areas proximal to tectonic features. (W. H. Al-Bazzaz, et al. 2006).

Structurally, the Mauddud Formation is dissected by a network of NE-SW-oriented normal faults associated with regional tectonic episodes during the Late Cretaceous and Tertiary (Alzankawi O., et al. 2014). These faults serve as both migration pathways and compartment boundaries, influencing fluid distribution and pressure communication. Seismic interpretation and structural modeling reveal fault throws ranging from a few meters to over 50 meters, with fracture density highest within 500 meters of major fault planes.

Reservoir characterization has integrated core, log, seismic, and production data to define the distribution of facies and petrophysical properties. The depositional environment ranges from mid-ramp peloidal grainstones to back-ramp mudstones, with high-energy shoal facies exhibiting the best reservoir quality. However, connectivity is limited without the presence of open fractures.

Evaluations in analog settings have demonstrated the consistent advantages of horizontal wells over vertical completions in tight carbonate intervals with thin oil columns. In gas injection zones, horizontal wells showed nearly double the productivity index of vertical. (Kohwar L. et Al. 2023)

Static modeling has delineated multiple flow units within the Mauddud, each with distinct porosity-permeability trends and capillary pressure characteristics. These flow units, in combination with the fracture network, control the productivity observed in existing wells. The STOIP estimation for the Mauddud reservoir in the Greater Burgan Field highlighted its strategic importance. The study underscored the value and critical aspect of proper planning development, drilling and completion strategy regarding the formation technical challenges.

Petrophysics

Petrophysical evaluation of the Mauddud Formation integrates core analysis, well logs, and production data to quantify reservoir quality and support static and dynamic modeling. Core measurements indicate an average porosity of 15-20% in the best reservoir intervals, with values locally exceeding 25% in high-energy grainstone facies. Permeability is generally low, ranging from less than 1 mD in tight mudstones to tens of millidarcies in dolomitized zones and grainstones, highlighting the heterogeneous nature of the reservoir.

Water saturation is variable, with lower values typically observed in structurally high areas and in the vicinity of open fractures. Resistivity logs, calibrated to core-derived Archie parameters, confirm the presence of hydrocarbon-bearing intervals, although the complex pore geometry in tight carbonates poses interpretation challenges. NMR logs have been used to refine the estimation of effective porosity and movable fluid volumes, identifying zones with higher producible oil potential.

Capillary pressure measurements reveal the presence of multiple pore systems, with microporosity dominating the matrix and macroporosity associated with fracture-enhanced flow units. Mercury injection capillary pressure (MICP) data indicate entry pressures consistent with low-permeability carbonates, and relative permeability curves suggest a strong dependence on fracture connectivity for sustained oil flow.

Petrophysical modeling has been incorporated into the static reservoir model to delineate flow units and guide well placement. Integration of log-derived properties with seismic attributes has improved the mapping of sweet spots, particularly in areas with favorable porosity-permeability trends and proximity to fault-related fracture corridors.

Geophysics

The geophysical framework for the Mauddud Formation development in the Greater Burgan Field relies heavily on the integration of 3D seismic interpretation, structural modeling, and seismic attribute analysis to characterize reservoir architecture and identify fracture-prone zones.

Seismic Data & Interpretation

High-quality, full-fold 3D seismic volumes acquired over the field were reprocessed to enhance signal-to-noise ratio, improve structural resolution, and provide better imaging of fault planes and thin carbonate layers. The seismic interpretation delineated a dense network of NE-SW-oriented faults, consistent with regional tectonic stress regimes, with vertical throws ranging from less than 5 m to more than 50 m. Fault proximity correlates strongly with higher fracture density and enhanced productivity.

Seismic Attributes & Fracture Detection

Seismic attribute analysis, including coherence, curvature, and spectral decomposition, was used to map subtle discontinuities and predict fracture intensity (Fig. 1). Amplitude anomalies and frequency attenuation patterns provided additional clues to lithological variations and fluid presence. These attributes were calibrated with wellbore image logs, which confirmed open fracture trends and azimuthal orientation.

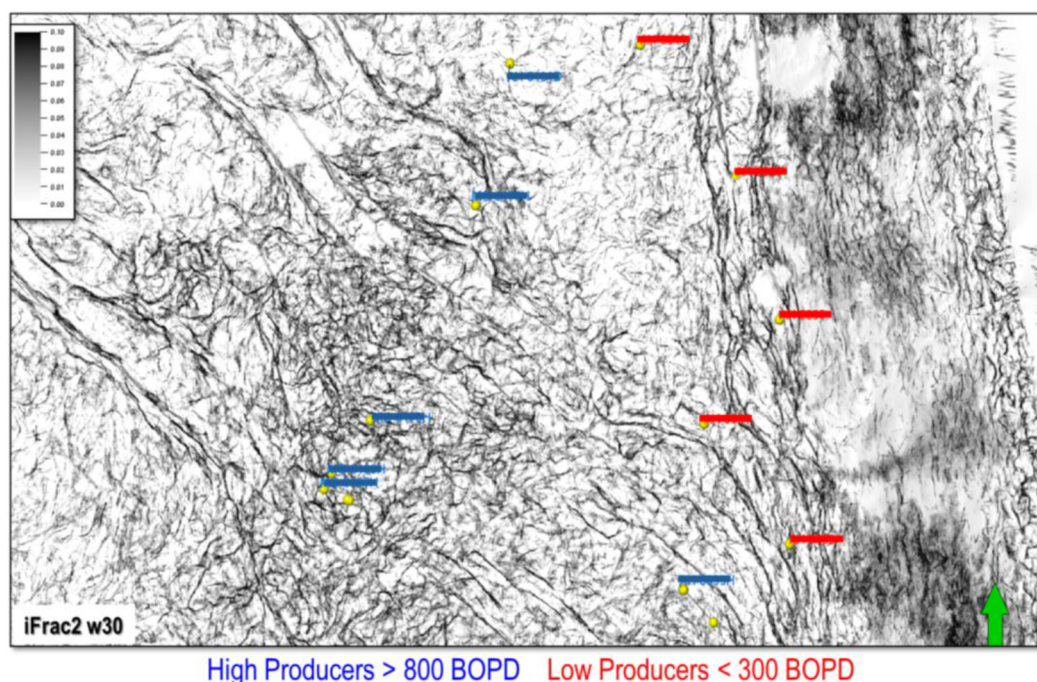


Figure 1—Map of regional discontinuities and fracture corridors resulting Seismic attribute iFrac

Structural Modeling

The interpreted horizons and faults were incorporated into a 3D structural model, providing the framework for subsequent static reservoir modeling and geomechanical simulation. Depth conversion was performed using check-shot and VSP data, ensuring consistency between seismic interpretation and well control. The

structural model revealed compartmentalization patterns that influence fluid distribution, pressure regimes, and optimal well trajectories.

Integration with Geomechanics & Reservoir Models

Seismic-derived fracture corridors were integrated with the 3D geomechanical model (MEM) to assess stress regimes and fracture reactivation potential under different operational scenarios. This integration guided the placement of horizontal wells and the design of multi-stage acid fracturing treatments, targeting structurally favorable zones with optimal stress orientation for stimulation efficiency.

Geomechanics

A comprehensive geomechanical evaluation was conducted to support well placement, stimulation design, and long-term reservoir management in the Mauddud Formation. The analysis integrated core-derived rock mechanical properties, wellbore image logs, drilling data, and 3D static and structural models to develop a calibrated MEM at both 1D (well-scale) and 3D (field-scale) resolutions.

Rock Mechanical Properties

Laboratory triaxial tests on core samples provided dynamic and static elastic properties, including Young's modulus, Poisson's ratio, and unconfined compressive strength (UCS). These measurements were correlated with well logs to establish continuous property profiles. The Mauddud carbonates display moderate stiffness and strength, with UCS values typically ranging from 50 to 120 MPa, depending on lithology and diagenetic overprint.

Stress Regime and Orientation

Borehole breakouts, drilling-induced tensile fractures, and borehole image logs were used to constrain the present-day stress regime. The Mauddud reservoir is characterized by a strike-slip to normal faulting regime, with maximum horizontal stress (SHmax) oriented approximately NE-SW, parallel to the main fracture and fault trends identified in seismic interpretation.

1D and 3D MEM Construction

The 1D MEMs were developed for key wells, incorporating vertical stress from density logs, pore pressure from MDT/RFT measurements, and elastic properties from log-core correlations. These were upscaled into a 3D MEM by integrating structural and stratigraphic frameworks from seismic interpretation. The 3D model enables spatial prediction of stress magnitudes and orientations across the field, highlighting areas prone to wellbore instability or fracture reactivation.

Applications for Drilling and Stimulation

Geomechanical outputs were used to optimize mud weight windows, casing design, and well trajectories to mitigate drilling risks such as losses and instability. The stress model also informed multi-stage acid fracturing design, ensuring that fracture propagation aligned with SHmax to maximize stimulated reservoir volume. Sensitivity analyses evaluated the potential for shear reactivation of existing fractures under varying drawdown and injection scenarios.

Integration with Reservoir Development

By coupling the 3D MEM with the Discrete Fracture Network (DFN) model, it was identified sweet spots where favorable stress conditions overlapped with high fracture density and reservoir quality. This integration was critical in selecting the location and completion strategy for the new horizontal wells that achieved a 60% production increase.

Discrete Fracture Network Modeling

A key component of the reservoir characterization in the Mauddud Formation was the development of a DFN model to accurately represent the spatial distribution, orientation, and connectivity of natural fractures. The DFN served as the foundation for understanding fracture-controlled permeability and optimizing well placement and stimulation design.

Data Sources and Calibration

The DFN construction relied on multiple datasets, including borehole image logs, core fracture descriptions, production logging data, and 3D seismic attribute volumes (coherence, curvature). Fracture orientations from image logs confirmed a dominant NE-SW trend, consistent with the maximum horizontal stress direction and major fault orientation in the field. Fracture aperture and spacing distributions were derived from core measurements and calibrated against well-test permeability data.

Fracture Classification

Fractures were categorized into open, partially open, and sealed based on image log resistivity contrasts and core observations. Open fractures, predominantly fault-related, showed the greatest impact on well productivity. Sealed fractures were often filled with calcite or anhydrite, contributing minimally to flow but influencing mechanical properties.

Model Construction

The 3D DFN was embedded within the static geological model, using stochastic methods constrained by fracture intensity maps derived from seismic attributes. Fault damage zones were explicitly modeled with higher fracture densities and connectivity. The DFN grid resolution was optimized to capture both large conductive fractures and smaller sub-seismic features without excessive computational cost.

Flow Simulation and Impact

Coupling the DFN with dynamic simulation demonstrated that matrix-only models underpredicted well performance in fractured areas. Dual-porosity/dual-permeability simulations provided a better match to historical production data, particularly for wells within 500 m of major faults. Sensitivity analyses indicated that fracture connectivity and orientation relative to wellbore trajectories significantly influenced stimulation effectiveness and production sustainability.

Integration with Development Planning

The DFN model was integrated with the 3D geomechanical model to identify "sweet spots" where fracture corridors aligned with favorable stress conditions, reducing the risk of fracture closure after stimulation. These insights guided the drilling of new horizontal wells and the design of multi-stage acid fracturing treatments, which resulted in a 60% production increase in targeted zones.

Methodology

The methodology adopted for the Mauddud Formation development was a fully integrated, multi-disciplinary workflow combining geology, petrophysics, geophysics, geomechanics, reservoir engineering, production engineering, and data science. The aim was to address the challenges of a low-permeability carbonate reservoir and achieve the strategic production target.

A comprehensive data gap analysis was conducted, reviewing geological, petrophysical, geophysical, and production datasets to identify key uncertainties in the reservoir description. To address these gaps, the team implemented targeted data acquisition, including new core samples, advanced wireline logs such as image logs, NMR, and MDT, as well as the reprocessing of legacy 3D seismic volumes. This reprocessing

improved imaging of thin beds, fault patterns, and subtle structural features, thereby supporting more accurate modeling inputs.

The static geological model was subsequently updated through refined stratigraphic correlations and detailed depositional facies mapping. A petrophysical re-interpretation was performed to capture diagenetic effects and define discrete flow units and internal baffles. These enhancements significantly improved the model's ability to represent heterogeneity, particularly in zones impacted by fracture corridors. Reprocessed 3D seismic data contributed to a higher resolution structural framework, which was further analyzed using seismic attributes- such as coherence, curvature, and spectral decomposition- to map fracture intensity and detect subtle discontinuities. These attributes served to constrain both the DFN and geomechanical models.

MEM were developed at both well and field scale to evaluate in-situ stresses, elastic properties, and fracture reactivation potential. These geomechanical outputs informed safe drilling design, wellbore stability assessments, and fracture stimulation planning. The DFN was built using a stochastic approach based on fracture data from image logs and core descriptions, and was calibrated against well-test permeability and production history. Fault damage zones were explicitly modeled to capture areas of higher fracture density and connectivity, as shown in Fig. 2.

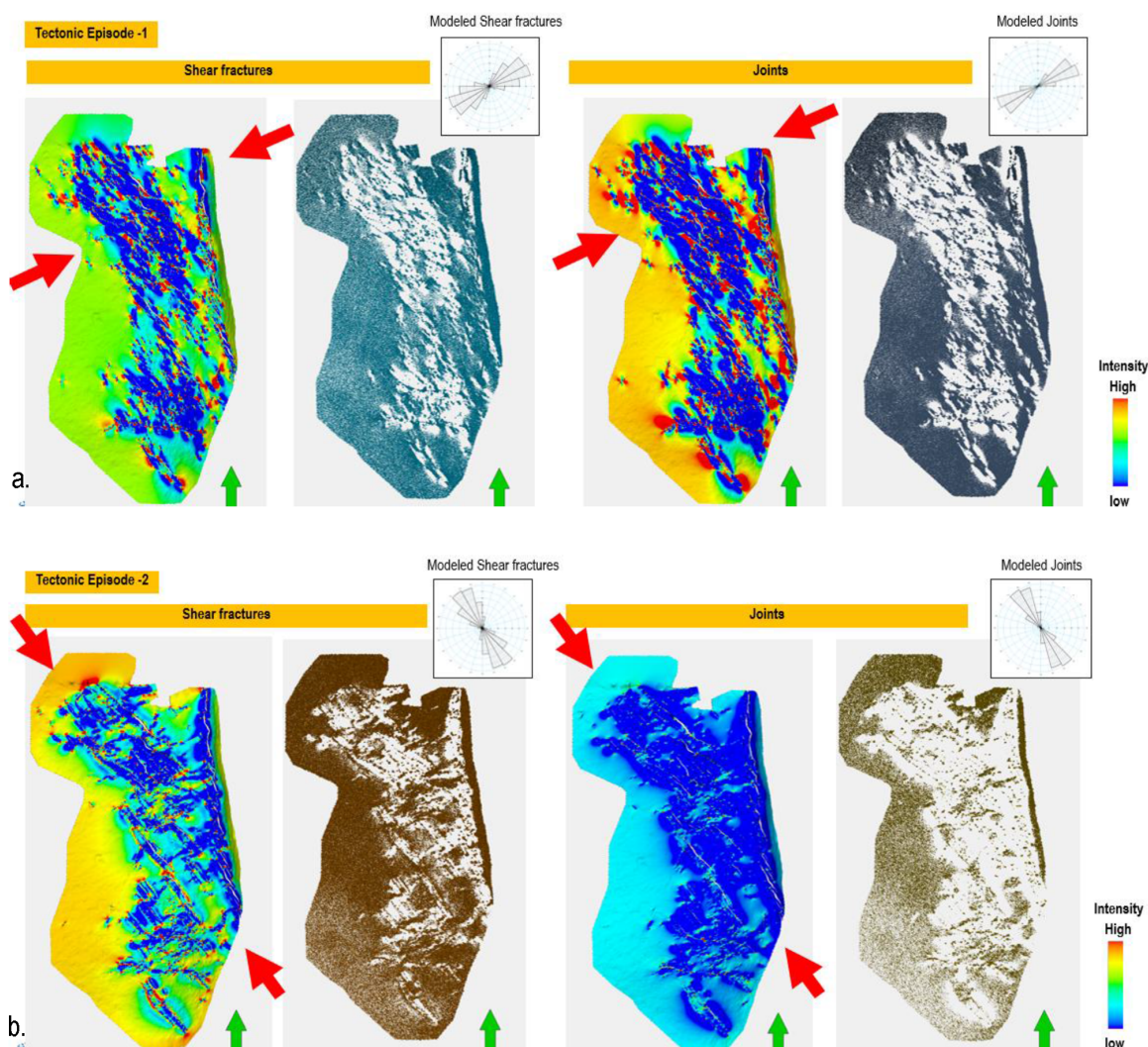


Figure 2—Modeling for different tectonic events by fracture drivers. a. tectonic Episode-1, b. tectonic Episode-2

Reservoir engineering analysis played a central role in integrating the updated static and dynamic models to better understand current field performance and to test forward-looking development strategies. Material

balance calculations and volumetric assessments refined the stock tank oil initially in place (STOIIP). Dual-porosity and dual-permeability simulations captured the distinct contributions from matrix and fracture networks. Sensitivity analyses were conducted to evaluate the impact of various strategies, including horizontal drilling and multi-stage acid fracturing, on recovery factor improvements. Additionally, pressure transient analysis (PTA) and rate transient analysis (RTA) were used to assess reservoir connectivity, fracture conductivity, and the extent of drainage area following stimulation efforts.

The production engineering evaluation focused on optimizing inflow and outflow performance. Inflow Performance Relationship (IPR) curves, nodal analysis, and wellbore hydraulics modeling were utilized to evaluate skin effects, optimize artificial lift strategies, and set operational parameters that minimize drawdown-induced formation damage while maximizing production. Candidate selection for workovers and new drills was informed by a combination of geological, reservoir, and production performance indicators. Criteria for candidate identification included proximity to high-fracture-density corridors, rock quality indices, historical stimulation effectiveness, and overall mechanical integrity of the wellbore. For new wells, geo-steered trajectories were planned to intersect sweet spots characterized by favorable stress orientation and fracture intensity.

Following intervention activities, productivity was assessed through comparative IPR analysis, as well as PTA and RTA interpretation. Production logging tools were employed to evaluate stimulation efficiency and to identify further optimization opportunities. The analysis confirmed that horizontal wells targeting high-fracture-density corridors delivered the most significant productivity gains, as illustrated in Fig. 3.

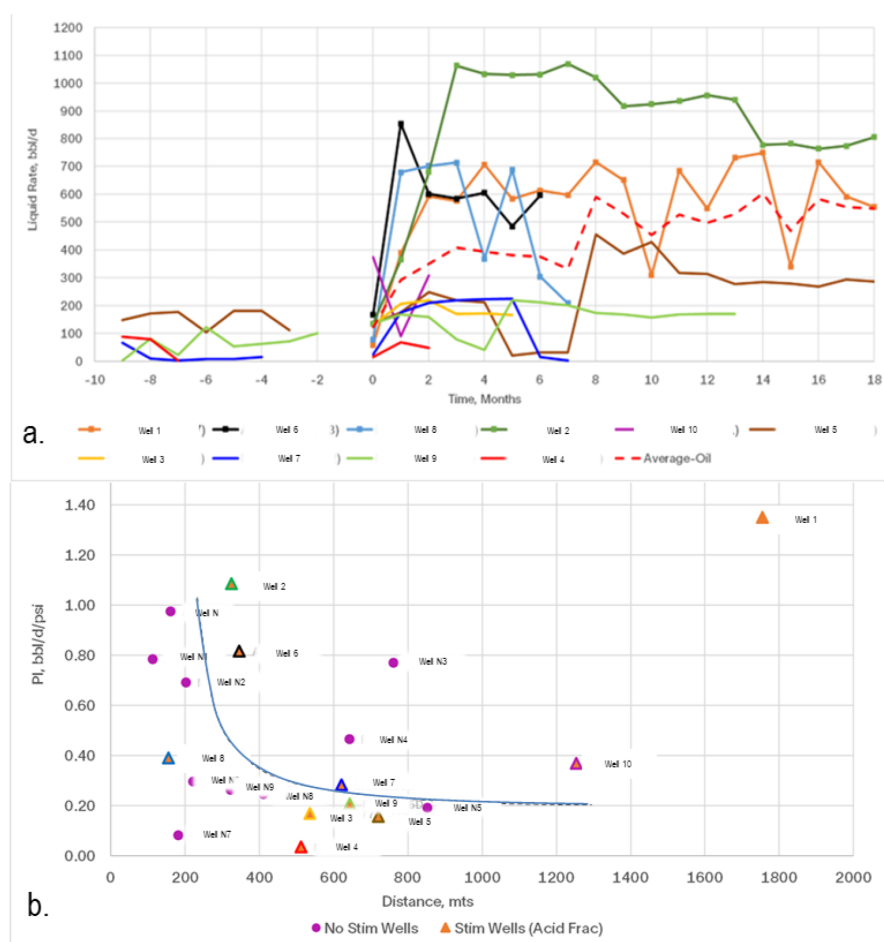


Figure 3—Productivity Analysis. a. vertical vs deviated wells (pre and post frac production); b. Productivity index vs distance to fault

Finally, the integrated workflow facilitated the deployment of advanced technologies such as geo-steered horizontal drilling, multistage acid stimulation, and enhanced zonal isolation methods. Validation of this approach came from two newly drilled horizontal wells, which achieved a sustained 60% increase in production, rising from 5 to 8 thousand barrels per day. This confirmed the predictive reliability of the integrated models. Continuous surveillance and iterative model updates were key to ensuring alignment between field development and the evolving behavior of the reservoir.

Results & Discussion

Static Model Outcomes

The updated static geological model integrated refined stratigraphic correlations, facies distributions, petrophysical re-interpretation, and discrete fracture network data including Key updates.

- **Facies Architecture:** High-energy shoal grainstones and dolomitized zones exhibited the best reservoir quality, with porosity up to 25% and permeability in the tens of millidarcies. Back-ramp mudstones and marl-rich intervals acted as internal flow barriers.
- **Flow Unit Delineation:** Multiple flow units were identified based on porosity-permeability trends and capillary pressure behavior. These units were explicitly modeled to capture heterogeneity in connectivity and productivity.
- **Fracture Integration:** Fracture density maps from seismic attributes and image logs were embedded into the static model, allowing spatial distribution of fracture corridors to be explicitly represented.
- **Model Validation:** History matching of dynamic simulations confirmed that incorporating fractures into the static model significantly improved the match with historical production trends compared to matrix-only models.

Reservoir Characterization Outcomes

The integrated reservoir characterization confirmed that the Mauddud Formation is a thinly layered, low-permeability carbonate reservoir with porosity averaging 15-20% in the best facies, but with significant heterogeneity driven by depositional and diagenetic processes. Seismic attribute analysis and image logs interpretation revealed that the dominant fracture orientation is NE-SW, aligning with the regional maximum horizontal stress direction. Faults and associated fracture corridors proved to be key contributors to enhanced productivity, particularly for wells located within 500 m of major structural features.

Geomechanics and DFN Insights

The 3D MEM identified a strike-slip to normal faulting stress regime, providing critical input for stimulation design. Integration of the MEM with the DFN demonstrated that high fracture connectivity coincides with favorable stress conditions, reducing the risk of fracture closure and improving stimulation efficiency. (Haas et al, 2023) This alignment was particularly valuable for planning horizontal well trajectories and fracture stage placement.

Reservoir Simulation and Performance Prediction

Dual-porosity/dual-permeability dynamic simulations provided a significantly better match to historical production data compared to single-porosity models. Simulations indicated that without fracture contribution, expected rates would be 30-50% lower. Sensitivity analyses showed that optimal production gains occur when horizontal wells are drilled parallel to SHmax and intersect multiple fracture corridors. Recovery factor improvements were most sensitive to stimulation effectiveness and drainage area expansion.

Production Engineering & Productivity Analysis

Nodal analysis and IPR evaluations pre- and post-intervention confirmed substantial productivity improvements. For the two newly drilled horizontal wells, multi-stage acid fracturing resulted in a sustained 60% increase in oil production, from approximately 5 Kbbbl/d to 8 Kbbbl/d. Post-stimulation PTA and RTA indicated increased fracture conductivity and expanded drainage radii, validating the design and confirming the predictive value of the integrated workflow.

Candidate Recognition & Workover Results

The candidate selection process identified wells with high remaining potential based on geological setting, proximity to faults, reservoir quality, and historical response to stimulation. Workovers performed on selected wells demonstrated production uplifts of 20-35%, further supporting the value of integrated screening.

Operational Learnings and Impact on Field Development Strategy

Wells targeting zones with high fracture density consistently demonstrated superior performance compared to those completed in matrix-dominated areas. It was observed that geomechanical constraints on fracture orientation played a critical role in sustaining the long-term benefits of stimulation treatments. The integration of seismic, geological, and production data into a unified modeling workflow significantly reduced subsurface uncertainty and improved the accuracy of well placement decisions.

These insights have had a direct impact on the field development strategy for the Mauddud Formation. The updated approach now prioritizes horizontal drilling within fracture-enhanced corridors, where reservoir connectivity is maximized. Multi-stage acid fracturing techniques are applied with consideration to the prevailing stress regime to ensure effective stimulation. In addition, the static model, DFN and MEM are continuously updated with new surveillance data, allowing for ongoing refinement of sweet spot mapping and dynamic reservoir understanding.

Conclusions & Recommendations

The updated static model incorporates facies architecture, petrophysical trends, and DFN data, resulting in a more accurate representation of reservoir heterogeneity and connectivity. The inclusion of fracture corridors significantly enhanced dynamic history matching compared to models that considered only the matrix.

Natural fractures, primarily oriented NE-SW and associated with faulting, have proven to be a major factor in driving productivity within the Mauddud Formation. Wells located within approximately 500 meters of major faults consistently outperform those situated in matrix-dominated zones.

The integration of geomechanics through a 3D Mechanical Earth Model has been essential for understanding stress regimes and optimizing wellbore stability, horizontal well trajectories, and stimulation stage placement. Aligning stimulation treatments with the direction of maximum horizontal stress has maximized fracture conductivity and long-term productivity.

Dynamic simulation using dual-porosity/dual-permeability models has provided more accurate predictions of production behavior by capturing both matrix and fracture contributions to flow. Sensitivity analysis revealed that stimulation effectiveness and fracture connectivity are the most influential factors in enhancing recovery.

Operationally, two new horizontal wells drilled and completed using the integrated workflow achieved a sustained 60% increase in production, rising from 5 to 8 Kbbbl/d. Additionally, workovers on selected candidate wells resulted in production uplifts of 20-35%, validating the selection criteria and methodology.

Overall, the integrated approach- combining geology, geophysics, petrophysics, geomechanics, reservoir engineering, and production engineering- has established a robust and repeatable workflow for unlocking the potential of the Mauddud Formation.

It is recommended to continue updating the static model, DFN, and MEM regularly with new drilling, logging, and production data. This will help refine sweet spot mapping and reduce uncertainty in well planning. Horizontal drilling should be prioritized in fracture- enhanced corridors, with well trajectories designed to intersect areas of maximum fracture density and aligned with the maximum horizontal stress (SHmax) direction.

Stimulation efforts should maintain multi-stage acid fracturing as the primary technique, with treatment volumes and stage spacing tailored to local stress conditions and fracture characteristics. Post-intervention surveillance and diagnostics should be expanded through the use of production logging, pressure transient analysis (PTA), and rate transient analysis (RTA) to assess stimulation effectiveness and support continuous improvement.

A systematic approach to candidate screening for workovers should be maintained, focusing on geological setting, reservoir quality, proximity to fracture corridors, and historical stimulation response. Additionally, the integration of emerging technologies- such as downhole monitoring tools and real-time fracture mapping- should be evaluated to further enhance stimulation placement and operational efficiency.

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